

```
In[*]:= Clear[pathHome, pathFGV, path]
pathHome = "G:\\Meu Drive\\ArtigoWikipedia\\PesquisaWikipedia\\";
pathFGV =
  "D:\\Users\\leandro.gorno\\Dropbox\\Research\\Projects\\2023-11 - Wikipedia
  graphs\\2025-07 - New data gathering\\Data\\";
path = pathHome;
```

```
In[*]:= Clear[graphPT3]
graphPT3 = Get[path <> "graphPT3.dat"]
```

Out[*]=

```
Graph[ Vertex count: 25 092
Edge count: 55 474]
```

```
In[*]:= Length[EdgeList[graphPT3]]
```

Out[*]=

55 474

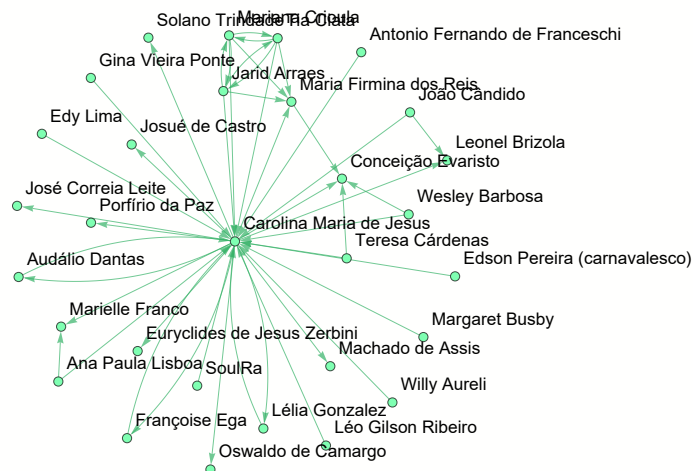
```
In[*]:= Length[VertexList[graphPT3]]
```

Out[*]=

25 092

```
In[*]:= NeighborhoodGraph[graphPT3, "Carolina Maria de Jesus", VertexLabels -> "Name"]
```

Out[*]=



Cómo puedo averiguar una lista de los vertex con mas centralidad en el grafo.

```
In[*]:= Part[VertexList[graphPT3], Ordering[VertexDegree[graphPT3], -20]]
```

Out[*]=

```
{Tom Jobim, Fernando Pessoa, Leonel Brizola, Nelson Rodrigues,
Jorge Amado, Machado de Assis, Madonna, Fernando Collor,
Gilberto Gil, Platão, José Sarney, Chico Buarque, Caetano Veloso,
Isabel II do Reino Unido, Jesus, Fernando Henrique Cardoso, Napoleão Bonaparte,
William Shakespeare, Getúlio Vargas, Luiz Inácio Lula da Silva}
```

Cuál es la diferencia entre Vertex Degree y Degree Centrality?

```
In[*]:= SortBy[{VertexList[graphPT3], DegreeCentrality[graphPT3]}^T, Last] // Reverse
```

VertexDegree

```

In[*]:= SortBy[{VertexList[graphPT3], DegreeCentrality[graphPT3]}, Last]

In[*]:= Part[VertexList[graphPT3], Ordering[EigenvectorCentrality[graphPT3], -50]]

In[*]:= SortBy[{VertexList[graphPT3], EigenvectorCentrality[graphPT3]}^T, Last] // Reverse

In[*]:= SortBy[{VertexList[graphPT3], BetweennessCentrality[graphPT3]}^T, Last] // Reverse

In[*]:= SortBy[{VertexList[graphPT3], ClosenessCentrality[graphPT3]}^T, Last] // Reverse

In[*]:= SortBy[{VertexList[graphPT3], ClosenessCentrality[graphPT3]}^T, Last] // Reverse

In[*]:= Part[VertexList[graphPT3], Ordering[EigenvectorCentrality[graphPT3], All, Greater]]

In[*]:= With[{c = ClosenessCentrality[graphPT3]}, Pick[VertexList[graphPT3], c, Max[c]]]

In[*]:= (*1. Your Gender API Function*) ClearAll[getGenero];
getGenero[titulo_String] := Module[{apiWiki, dadosWiki, pagina, qid,
  apiWD, dadosWD, claims, generoQ, apiLabel, dadosLabel}, Quiet[apiWiki =
  "https://pt.wikipedia.org/w/api.php?action=query&prop=pageprops&titles=" <>
  URLEncode[titulo] <> "&format=json";
  dadosWiki = Import[apiWiki, "RawJSON"];
  pagina = First@Values@dadosWiki["query", "pages"];
  qid = pagina["pageprops", "wikibase_item"];
  If[qid === Missing["KeyAbsent", _], Return[None]];
  apiWD = "https://www.wikidata.org/w/api.php?action=wbgetentities&ids=" <>
  qid <> "&props=claims&format=json";
  dadosWD = Import[apiWD, "RawJSON"];
  claims = dadosWD["entities", qid, "claims"];
  If[! KeyExistsQ[claims, "P21"], Return[None]];
  generoQ = claims["P21"][[1, "mainsnak", "datavalue", "value", "id"]];
  apiLabel = "https://www.wikidata.org/w/api.php?action=wbgetentities&ids=" <>
  generoQ <> "&props=labels&languages=pt&format=json";
  dadosLabel = Import[apiLabel, "RawJSON"];
  dadosLabel["entities", generoQ, "labels", "pt", "value"]]]

(*2. Extract the raw Top 20 data first*)
getRawTop20[metric_] := Take[ReverseSortBy[
  Transpose[{VertexList[graphPT3], N[metric[graphPT3]]}], Last], UpTo[20]];

rawDeg = getRawTop20[DegreeCentrality];
rawBet = getRawTop20[BetweennessCentrality];
rawEig = getRawTop20[EigenvectorCentrality];

(*3. Find unique articles and fetch their
  genders ONCE (to save time and avoid API bans)*)
Print["Consultando Wikidata... isso pode levar alguns segundos."];
uniqueArticles = Union[Join[rawDeg[[All, 1]], rawBet[[All, 1]], rawEig[[All, 1]]];
genderDictionary = AssociationMap[getGenero, uniqueArticles];

(*4. Row Formatter:Applies color based on the dictionary and rounds decimals*)
formatRow[{article_, score_}] := {If[genderDictionary[article] === "feminino",
  Style[article, Bold, Darker[Red]], article], Round[score, 0.001]};

```

(*5. Apply the formatting to our raw lists*)

```
topDeg = TableForm[Map[formatRow, rawDeg], TableAlignments → {Left, Right}];
topBet = TableForm[Map[formatRow, rawBet], TableAlignments → {Left, Right}];
topEig = TableForm[Map[formatRow, rawEig], TableAlignments → {Left, Right}];
```

(*6. Build the Grid, forcing exactly equal widths*)

```
Tablacentralidad =
Grid[{{Item[Style["Centralidad de Grado", Bold], ItemSize → Scaled[1 / 3],
  Alignment → Center], Item[Style["Centralidad de Intermediação", Bold],
  ItemSize → Scaled[1 / 3], Alignment → Center],
  Item[Style["Centralidad de vector propio", Bold], ItemSize → Scaled[1 / 3],
  Alignment → Center]}}, {Item[topDeg, Alignment → Center],
  Item[topBet, Alignment → Center], Item[topEig, Alignment → Center]}}, Frame → All,
Background → {{Lighter[Blue, 0.95], Lighter[Green, 0.95], Lighter[Orange, 0.95]},
{Lighter[Gray, 0.8], None}}, Alignment → Top]
```

Consultando Wikidata... isso pode levar alguns segundos.

Out[*]=

Centralidade de Grau	Centralidade de Intermediação	Centralidade de vetor propio	
	João Vargas	2.0518	
Luiz Inácio Lula da Silva	1.5284	Cúlio Vargas	0.0
Cúlio Vargas	1.4468	Luiz Inácio Lula da Silva	0.0
William Shakespeare	1.4090	João Buarque	0.0
Napoleão Bonaparte	1.0734	Manoel Henrique Cardoso	0.0
Manoel Henrique Cardoso	1.0138	João Veloso	0.0
João Veloso	1.0073	João Buarque	0.0
João Buarque	0.9533	João Sarney	0.0
João Sarney	0.9440	Manoel Collor	0.0
Manoel Collor	0.9097	João Brizola	0.0
João Brizola	0.8914	João de Moraes	0.0
João de Moraes	0.8491	João Bethânia	0.0
João Bethânia	0.7543	João Jobim	0.0
João Jobim	0.7226	João Amado	0.0
João Amado	0.6281	João Kubitschek	0.0
João Kubitschek	0.6391	João Gilberto	0.0
João Gilberto	0.5877	João de Andrade	0.0
João de Andrade	0.5774	João Serra	0.0
João Serra	0.5705	João Neves	0.0
João Neves	0.5494	João Carlos Magalhães	0.0
João Carlos Magalhães	0.5299	João de Andrade	0.0
João de Andrade	0.5277		

```
In[*]:= Export[path <> "centralidad.png", Tablacentralidad]
```

Out[*]=

G:\Meu Drive\ArtigoWikipedia\PesquisaWikipedia\centralidad.png

```
In[*]:= writersgraph =
CommunityGraphPlot[graph, FindGraphCommunities[graph, Method → "Modularity"]]
```

```
In[*]:= Export[path <> "writersgraph.png", writersgraph]
```

Out[*]=

G:\Meu Drive\ArtigoWikipedia\PesquisaWikipedia\writersgraph.png

```
In[*]:= CommunityGraphPlot[graph, FindGraphCommunities[graph, Method → "Centrality"]]
```

```

In[*]:= CommunityGraphPlot[graph, FindGraphCommunities[graph, Method → "Hierarchical"]]

In[*]:= Length /@ FindGraphCommunities[graph]

Out[*]:=
{270, 203, 185, 117, 59, 26, 14, 5, 3, 2}

In[*]:= graph = NeighborhoodGraph[graphPT3,
  writers = {"Carolina Maria de Jesus", "Clarice Lispector", "Gabriela Mistral",
    "Júlia Lopes de Almeida", "Machado de Assis", "Lima Barreto",
    "Mário de Andrade", "Jorge Luis Borges"}, VertexLabels → "Name", EdgeStyle → Thin,
    EdgeShapeFunction → {"Arrow", "ArrowSize" → .008}}, ImageSize → 1000]

In[*]:= graphCL = NeighborhoodGraph[graphPT3,
  writers = {"Carolina Maria de Jesus", "Lima Barreto (escritor)"},
  VertexLabels → "Name", EdgeStyle → Thin,
  EdgeShapeFunction → {"Arrow", "ArrowSize" → .008}}, ImageSize → 1000]

In[*]:= CarolinaLima =
  CommunityGraphPlot[graph, FindGraphCommunities[graph, Method → "Modularity"]]

In[*]:= graph = NeighborhoodGraph[graphPT3,
  writers = {"Jorge Luis Borges", "Gabriela Mistral"}, VertexLabels → "Name",
  EdgeStyle → Thin, EdgeShapeFunction → {"Arrow", "ArrowSize" → .008}}, ImageSize → 1000]

In[*]:= CommunityGraphPlot[graph, FindGraphCommunities[graph, Method → "Modularity"]]

In[*]:= graph = NeighborhoodGraph[graphPT3,
  writers = {"Clarice Lispector", "Machado de Assis"}, VertexLabels → "Name",
  EdgeStyle → Thin, EdgeShapeFunction → {"Arrow", "ArrowSize" → .008}}, ImageSize → 1000]

In[*]:= CommunityGraphPlot[graph, FindGraphCommunities[graph, Method → "Modularity"]]

In[*]:= graph = NeighborhoodGraph[graphPT3,
  writers = {"Mário de Andrade", "Júlia Lopes de Almeida"}, VertexLabels → "Name",
  EdgeStyle → Thin, EdgeShapeFunction → {"Arrow", "ArrowSize" → .008}}, ImageSize → 1000]

In[*]:= CommunityGraphPlot[graph, FindGraphCommunities[graph, Method → "Modularity"]]

In[*]:= graphGM = NeighborhoodGraph[graphPT3,
  writers = {"Gabriela Mistral"}, VertexLabels → "Name", EdgeStyle → Thin,
  EdgeShapeFunction → {"Arrow", "ArrowSize" → .008}}, ImageSize → 1000]

In[*]:= Clear[cleangraph, cleanededges, inlinks, outlinks, centrality, de, dv, target]
target = "Gabriela Mistral";
cleangraph = graphPT3;
cleanedges = EdgeList[cleangraph];
inlinks =
  Inner[Rule, VertexList[cleangraph], DegreeCentrality[cleangraph, "In"], List];
outlinks =
  Inner[Rule, VertexList[cleangraph], DegreeCentrality[cleangraph, "Out"], List];
centrality = Inner[Rule, VertexList[cleangraph], DegreeCentrality[cleangraph], List];
(* Centrality used to select *)
de = Select[EdgeList[cleangraph],
  Or[MatchQ[#, DirectedEdge[target, _]], MatchQ[#, DirectedEdge[_ , target]]] &];
dv = Union[Flatten[Map[List @@ # &, de]]];

In[*]:= dv

```

```

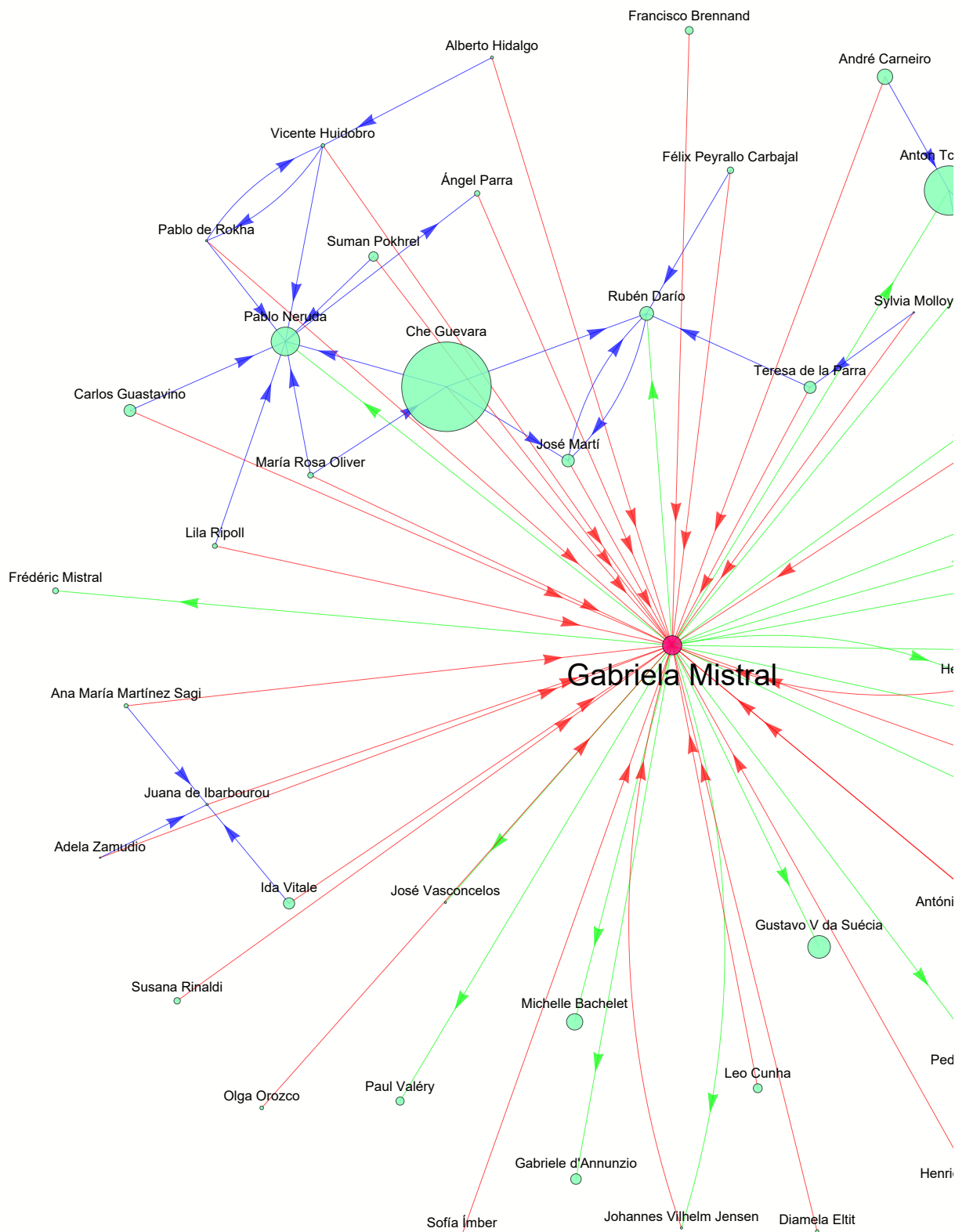
In[*]:= Clear[smalldges, smallgraph, vc]
smalldges =
  Select[cleanedges, And[MemberQ[dv, #[[1]], MemberQ[dv, #[[2]], #[[1]] != #[[2]]] &];
smallgraph = Graph[smalldges, ImageSize -> 800, VertexLabels -> "Name"];
vc = Rescale[Ordering[VertexList[smallgraph] /. centrality]] // N;

In[*]:= Clear[centralvertex, vin, vout, vertexsize, centraledges, centralgraph]
{centralvertex, vin, vout} =
  Transpose[Take[Reverse[SortBy[Map[{#, # /. inlinks, # /. outlinks, # /. centrality} &,
    Union[Flatten[Apply[List, smalldges, 2]]]],
    #[[4]] &]], Min[50, Length[vc]]]]][[1 ;; 3]];
If[MemberQ[centralvertex, target],
  Print[target <> " is already in!"], PrependTo[centralvertex, target];
PrependTo[vin, target /. inlinks];
PrependTo[vout, target /. outlinks];
Print[target <> " added!"]]
vertexsize = 1.3 (0.6 Rescale[vin] + 0.4 Rescale[vout]);
centraledges = Select[EdgeList[cleangraph],
  And[MemberQ[centralvertex, #[[1]], MemberQ[centralvertex, #[[2]], #[[1]] != #[[2]]] &];
centralgraph = Graph[centraledges, VertexSize -> Thread[centralvertex -> vertexsize],
  VertexStyle -> {target -> Orange, Opacity[0.8]},
  VertexLabels -> {target -> Placed[Style[target, 20], Below], Placed["Name", Above]},
  EdgeStyle -> {target -> _ -> Blue, _ -> target -> Red, Green},
  EdgeShapeFunction -> {{"Arrow", "ArrowSize" -> 0.015}}, ImageSize -> 900]

Gabriela Mistral is already in!

```

Out[]:=



In[]:= **Export**[path <> "graphGM.png", %]

Out[]:=

G:\Meu Drive\ArtigoWikipedia\PesquisaWikipedia\graphGM.png

In[]:= **Clear**[cleangraph, cleanedges, inlinks, outlinks, centrality, de, dv, target]
target = "Jorge Luis Borges";

```

cleangraph = graphPT3;
cleanedges = EdgeList[cleangraph];

inlinks = Inner[Rule, VertexList[cleangraph], DegreeCentrality[cleangraph, "In"], List];
outlinks =
  Inner[Rule, VertexList[cleangraph], DegreeCentrality[cleangraph, "Out"], List];
centrality = Inner[Rule, VertexList[cleangraph], DegreeCentrality[cleangraph], List];

de = Select[EdgeList[cleangraph],
  Or[MatchQ[#, DirectedEdge[target, _]], MatchQ[#, DirectedEdge[_ , target]]] &];
dv = Union[Flatten[Map[List @@ # &, de]]];

Clear[smalldges, smallgraph, vc]
smalldges =
  Select[cleanedges, And[MemberQ[dv, #[[1]]], MemberQ[dv, #[[2]]], #[[1]] != #[[2]]] &];
smallgraph = Graph[smalldges, ImageSize -> 800, VertexLabels -> "Name"];
vc = Rescale[Ordering[VertexList[smallgraph] /. centrality]] / N;

Clear[centralvertex, vin, vout, vertexsize,
  centraledges, centralgraph, genderDict, vertexColors]
{centralvertex, vin, vout} =
  Transpose[Take[Reverse[SortBy[Map[{#, # /. inlinks, # /. outlinks, # /. centrality} &,
    Union[Flatten[Apply[List, smalldges, 2]]],
    #[[4]] &]], Min[50, Length[vc]]]]][[1 ;; 3];

If[MemberQ[centralvertex, target],
  Print[target <> " is already in!"], PrependTo[centralvertex, target];
  PrependTo[vin, target /. inlinks];
  PrependTo[vout, target /. outlinks]; Print[target <> " added!"]

(*Your exact sizing math*)
vertexsize = 1.3 (0.6 Rescale[vin] + 0.4 Rescale[vout]);

centraledges = Select[EdgeList[cleangraph],
  And[MemberQ[centralvertex, #[[1]]], MemberQ[centralvertex, #[[2]]], #[[1]] != #[[2]]] &];

(* ===== *)
(*UPGRADED GENDER API FUNCTION*)
(* ===== *)
ClearAll[getGenero];
getGenero[titulo_String] := Module[{apiWiki, dadosWiki, pagina, qid,
  apiWD, dadosWD, claims, generoQ, apiLabel, dadosLabel, result}, Pause[0.2];
  (*CRITICAL:Prevents Wikidata from blocking you for too many requests*)
  Quiet[(*Replace underscores with spaces just in case*)apiWiki =
    "https://pt.wikipedia.org/w/api.php?action=query&prop=pageprops&titles=" <>
    URLEncode[StringReplace[titulo, "_" -> " "] <> "&format=json";
    dadosWiki = Import[apiWiki, "RawJSON"];
    pagina = First@Values@dadosWiki["query", "pages"];
    qid = pagina["pageprops", "wikibase_item"];
    If[MissingQ[qid], Return["none"]];

```

```

apiWD = "https://www.wikidata.org/w/api.php?action=wbgetentities&ids=" <>
  qid <> "&props=claims&format=json";
dadosWD = Import[apiWD, "RawJSON"];
claims = dadosWD["entities", qid, "claims"];
If[! KeyExistsQ[claims, "P21"], Return["none"]];
generoQ = claims["P21"][[1, "mainsnak", "datavalue", "value", "id"]];
apiLabel = "https://www.wikidata.org/w/api.php?action=wbgetentities&ids=" <>
  generoQ <> "&props=labels&languages=pt&format=json";
dadosLabel = Import[apiLabel, "RawJSON"];
result = dadosLabel["entities", generoQ, "labels", "pt", "value"];
(*Convert to lowercase to ensure perfect matching*)
If[StringQ[result], ToLowerCase[result], "none"]]]

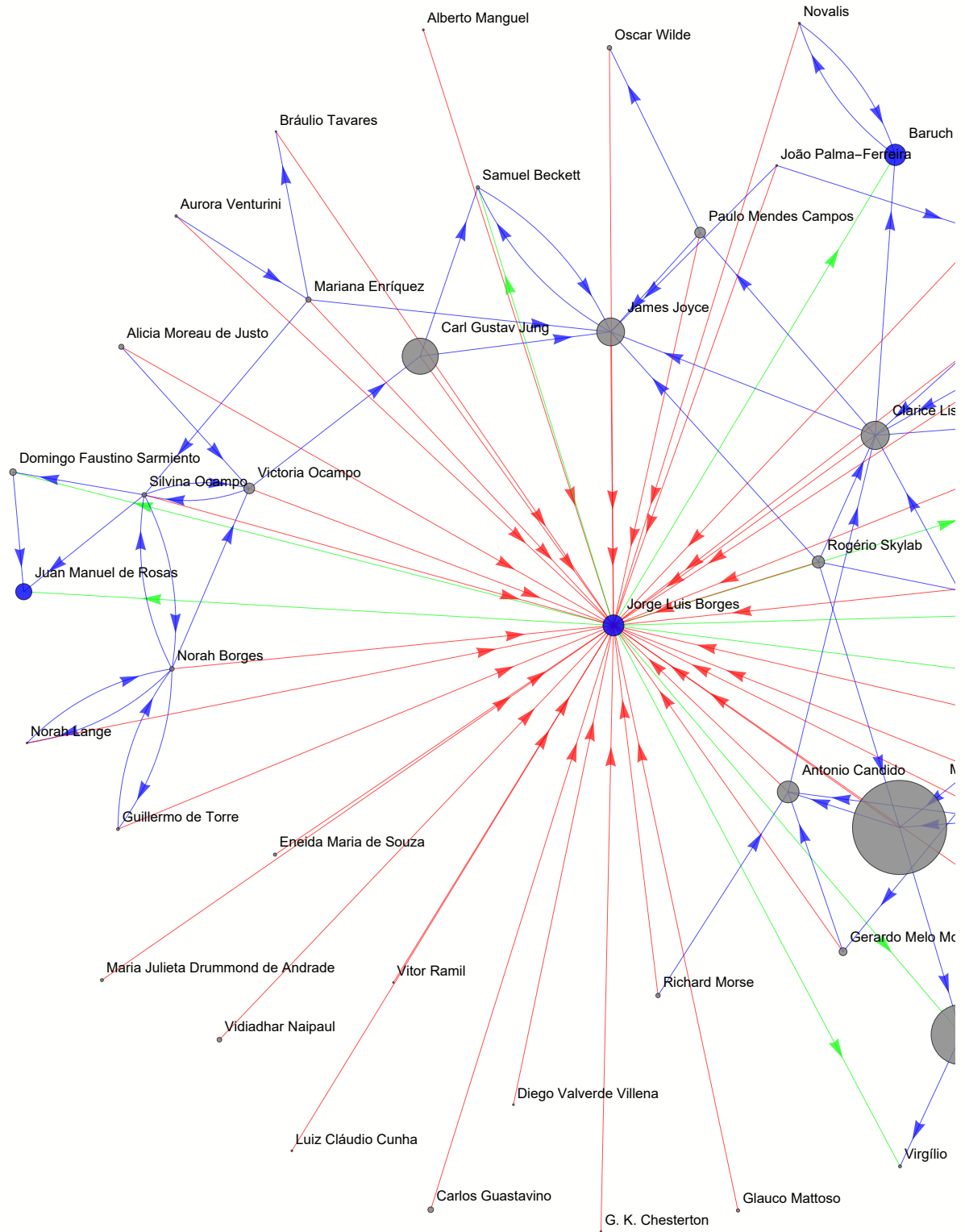
(*Fetch Genders*)
Print["Consultando Wikidata... (Isso levará ~10 segundos)"];
genderDict = AssociationMap[getGenero, DeleteCases[centralvertex, target]];

(*Upgraded matching logic using StringContainsQ to catch variations*)
vertexColors = Map[Function[{v}, Module[{gen = ToString[genderDict[v]]},
  v → Which[v == target, Directive[Green, Opacity[0.8]], StringContainsQ[
    gen, "feminino", IgnoreCase → True], Directive[Orange, Opacity[0.8]],
    StringContainsQ[gen, "masculino", IgnoreCase → True], Directive[Green,
    Opacity[0.8]], True, Directive[Gray, Opacity[0.8]]]]], centralvertex];

(* ===== *)
(*Build Graph*)
(* ===== *)
centralgraph = Graph[centraledges,
  VertexSize → Thread[centralvertex → vertexsize], VertexStyle → vertexColors,
  VertexLabels → {target → Placed[Style[target, 20], Below], _ → "Name"},
  EdgeStyle → {target ↔ _ → Blue, _ ↔ target → Red, Green},
  EdgeShapeFunction → {"Arrow", "ArrowSize" → 0.015}}, ImageSize → 900]
Jorge Luis Borges is already in!
Consultando Wikidata... (Isso levará ~10 segundos)

```

Out[]=



Export[path <> "centralmario.png", centralgraph, ImageResolution -> 500]

Out[]=

G:\Meu Drive\ArtigoWikipedia\PesquisaWikipedia\centralborges.png